Reflective Journal – Module 6: Regression and Classification Models

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1. Introduction

In this lab, I learned about the difference between regression and classification in machine learning. I used the Titanic dataset to build two models: Linear Regression to predict passenger fares and Logistic Regression to predict whether passengers survived. This lab helped me understand how machine learning models use data to make predictions and why it is important to evaluate their results.

2. Description of the Experience

During the lab, I used Python and Pandas to load and prepare the Titanic dataset. For Linear Regression, I used passenger age and class to predict fare prices. I trained the model, made predictions, and used Mean Squared Error (MSE) to measure how far the predicted fares were from the actual fares.

For Logistic Regression, I used age, passenger class, and sex to predict survival. I had to convert the sex values into numbers so the model could work with them. After training and testing the model, I used accuracy to evaluate its performance.

3. Personal Reflection and Learning

At first, I thought regression and classification were almost the same because both used data to make predictions. After completing this lab, I understood that the main difference is what the model predicts. Regression predicts a number, while classification predicts a category. Seeing how both models worked on the same Titanic dataset helped me better understand this difference.

4. Challenges, Improvements, and Questions

One challenge I faced was understanding which evaluation metric to use for each model. I learned that MSE measures the difference between predicted and actual numerical values, while accuracy measures how many classifications were correct. This helped me realize that choosing the right evaluation method is just as important as building the model.

In the future, I want to practice testing different features and comparing models to see whether I can improve their performance. I also wonder whether using additional information, such as the number of family members aboard or the passenger's fare, could help predict survival more accurately. I would also like to learn how to identify which features have the greatest influence on a model's predictions.

5. Conclusion

Overall, this lab helped me understand how regression and classification models work and how to evaluate their performance. I learned that building a model involves more than writing code; I also need to prepare the data, select the correct features, choose an appropriate evaluation metric, and interpret the results. My biggest takeaway is that machine learning requires understanding what the results mean, recognizing the limitations of a model, and looking for ways to improve it. I feel more confident using Python to build basic machine learning models and want to continue developing these skills in future labs.